MILITARY SPECIFICATION

DISPENSER CONTROL PANEL FOR M56/XM58, MINES/MARKERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection and tests for the dispenser control panel used in the Mines, Markers, Dispensing, Subsystem, Aircraft: M56/XM58.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 -	Quality Assurance Terms and Definitions
MIL-STD-810 -	Military Standard Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-1168-	Lot Numbering of Ammunition
MIL-STD-1235-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1345

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DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKAGING DRAWINGS

9287814 - M56/XM58 Dispenser Control panel
9287747 - Packing and markings for drum, metal, shipping for assembly, dispenser control panel and aircraft wiring harness

INSPECTION EQUIPMENT DRAWINGS

9278652 - Solenoid Test load.

(Copies of specifications; standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be In accordance with the applicable drawings and specifications.

3.2 Assembly. - The control panel shall comply with all requirements specified on drawing (dwg.) no. 9287814, as applicable, all associated drawings and with all requirements specified in applicable specifications

3.3 Performance. - Unless otherwise specified, all tests on the control panel shall be performed at ambient temperature.

3.4 Helicopter Vibration. - The control panel assembly shall perform all functions satisfactorily after being subjected to Test Method 514, Curve M of MIL-STD-810

3.5 Rear Panel Connector Ground Continuity. - The maximum resistance between all ground connected pins shall be fifty (50) milliohms.

3.6 Firing, Mines-Pairs. - The control panel shall fire in all combination modes as indicated by Tables A, B and D of dwg. 9288645 in accordance with the output requirements described by Curve A.

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3.7 Firing, Mines-Singles. - The control panel shall fire in all combination modes as indicated by Tables A, B and D of dwg. 9288645 in accordance with the output requirements described by Curve B.

3.8 Firing, Markers - Pairs. - The control panel shall fire in all combination modes as indicated by Tables A, B and D of dwg. 9288645 in accordance with the output requirements described by Curve C.

3.9 Firing, Markers-Singles. - The control panel shall fire in all combination modes as indicated by Tables A, B and D of dwg. 9288645 in accordance with the output requirements described by Curve D.

3.10 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All electrical assemblies shall be free of harmful cleaning and soldering agents that may be Injurious to the electrical assembly. All marking shall be neat and sharply defined.

3.11 Submission of results of contractor-conducted examinations and tests. - The contractor shall forward copies of the results of any of his examinations or tests to the designated address as specified in 6.4.

3.12 First article inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified In the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection re-

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quirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product .-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following Information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified In this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced In any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection -

4.2.1 Submission .-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of:

a. Five (5) sets of all sub-assemblies
b. Two (2) complete control panel assemblies

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All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. -Assemblies and components will be subjected by the Government to the examinations and tests as follows:

Five (5) sets of sub-assemblies shall be inspected for the requirernents of 4.3.2.1 thru 4.3.2.9.

b. The two (2) complete control panels shall be:

1. Inspected for the requirements of 4.3.2.10.
2. Tested for the vibration test 4.4.1
3. Tested for the requirements of 4.3.3.1 thru 4.3.3.8.

4.2.3 Rejection. - If any component, sub-assembly or assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component, sub-assembly or assembly to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term “inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities pre-

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sented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified In MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination .-The examinations listed in the following subparagraphs shall be performed on Inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, It shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection Is specified, and be documented as part of the contractor's Inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Printed Wiring, Board "A" Assembly (see dwg. 9287761 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	**None defined**		
Major:	**AQL 0.65 percent**		
101.	Loose, missing or improperly positioned components.....	Visual/ Manual	01001

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102.	Conformed coating on connector pins	Visual	01002
103.	Rivet improperly secured	Visual	01003
104.	Improper soldering	Visual	01004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	01005

4.3.2.2 Printed Wiring, Board "B" Assembly (see dwg. 9287763 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose, missing or improperly positioned components	Visual/ Manual	02001
102.	Conformed coating on connector pins	Visual	02002
103.	Rivet improperly secured	Visual	02003
104.	Improper soldering	Visual	02004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	02005

4.3.2.3 Printed Wiring, Board "C" Assembly (see dwg. 9287765 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose missing or improperly positioned components	Visual/ Manual	03001
102.	Conformed coating on connector pins	Visual	03002
103.	Rivet improperly secured	Visual	03003
104.	Improper soldering	Visual	03004

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	03005

4.3.2.4 Printed Wiring, Board "D" Assembly (see dwg. 9287767 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose, missing, or improperly positioned components	Visual/ Manual	04001
102.	Conformal coating on connector pins	Visual	04002
103.	Rivet improperly secured	Visual	04003
104.	Improper soldering	Visual	04004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	04005

4.3.2.5 Printed Wiring, Board "E" Assembly (see dwg. 9287769 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose or missing or improperly positioned components	Visual Manual	05001
102.	Conformal coating on connector pins	Visual	05002
103.	Improper wiring	Visual	05003
104.	Improper soldering	Visual	05004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	05005

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4.3.2.6 Front Panel Assembly (see dwg. 9283024 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Loose or missing or improperly positioned components	Visual/ Manual	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	06002

4.3.2.7 Bracket Panel Assembly (see dwg. 9287751 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Loose or missing or improperly positioned components	Visual/ Manual	07001
Minor;	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	07002

4.3.2.8 Rear Panel, Left and Right Side Panels Assemblies (see dwg. 9283028, 9287824 and 9287823 covering a detail of dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Loose or missing or improperly positioned components	Visual/ Manual	08001

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	08002

4.3.2.9 M56/XM58 Dispenser Control Panel (Prior to inserting printed circuit cards A, B, C, D, and E; front, bracket, sides and rear panels. See dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose or missing or improperly positioned components	Visual/ Manual	09001
102.	Brackets and panels loose	Manual	09002
103.	Improper wiring	Visual	09003
104.	Improper soldering	Visual	09004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	09005

4.3.2.10 M56/XM58 Dispenser Control Panel (After securing all boards and panels) (see dwg. 9287814).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Loose, or missing components..	Visual/ Manual	10001
102.	Improper captions	Visual	10002
103.	Paint incomplete or damaged ..	Visual	10003
104.	Improper soldering	Visual	10004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.10)	Visual	10005

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4.3.3 TESTING

4.3.3.1 Rear Panel connector ground continuity (see 3.5).- Major Defect - Code No. 07001. This test shall be performed 100 percent in accordance with 4.4.2. Any control panel failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test)

4.3.3.2 Jettison Power (see dwg. 9287814). - Major Defect - Code No. 07001. - This test shall be performed 100 percent in accordance with 4.4.3. Any control panel failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.3.3 Control Panel Assembly Light Dimming. - (see dwg. 9287814) - Major defect - Code No. 07003. This test shall be performed 100 percent in accordance with 4.4.4. Any panel failing to comply with all Of the requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.3.4 Press-to-test switch lighting. - (see dwg. 9287814) Major Defect - Code No. 07004. This test shall be performed 100 percent in accordance with 4.4.5. any control panel classed defective shall be classed defective and removed from the lot. (non-destructive test).

4.3.3.5 Firing, Mines-Pairs (see 3.6). - Major Defect - Code No. 07005. Each control panel assembly, after having successfully passed the test requirements of 4.4.3.1, shall be tested in accordance with 4.4.6. Any control panel assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.3.6 Firing, Mines-Singles (see 3.7) - Major Defect - Code No. 07006. Each control panel assembly, after having successfully passed the firing, mines - pairs test requirements of 4.3.3.6, shall be tested in accordance with 4.4.7. Any control panel assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.3.7 Firing, Markers-pairs (see 3.8). - Major Defect - Code No. 07007. Each control panel assembly, after having successfully passed the mines-singles firing test requirements

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of 4.3.3.7, shall be tested in accordance with 4.4.8. Any control panel assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.3.8 Firing, Markers-Singles (see 3.9) - Major Defect, Code No. 07008. Each control panel assembly, after having successfully passed the markers-pairs firing test requirements of 4.3.3.8, shall be tested in accordance with 4.4.9. Any control panel assembly failing to comply with all of the specified requirements shall be classed defective and removed from the lot. (Non-destructive test).

4.3.4 Inspection Equipment.-Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.4 Test Methods and Procedures

4.4.1 Helicopter Vibration. - The control panel shall be tested in accordance with MIL-STD-810, Method 514, Curve M.

4.4.2 Rear Panel Connector Ground Continuity. - The resistance between all ground connected pins shall be measured using an ohmmeter. Any sample failing to conform to the specified requirement shall be classed defective.

4.4.3 Jettison Power. - Twenty-eight (28) minus four (4) (28.0 -4.0) v.d.c. shall be applied with the positive (+) polarity to the “Jettison Power”, pin - K. Switch, S5FP, “Jettison” shall be thrown and the input voltage (28.0 - 4.0 v.d.c.) shall be measured at the output pins for ”Left Jettison”, Pin - J and "Right Jettison”, Pin - H in the rear panel connector, PIRP of dwg. 9287814. Any sample failing to conform to the applicable requirement shall be classed defective.

4.4.4 Control Panel Assembly Light Dimming. - Twenty-eight (28) v.d.c. shall be applied with the positive (+) polarity to the "+28V Panel Lighting”, Pin - N in the rear panel connector P1RP of dwg. 9287814. Switch shall be turned all the way in the clockwise direction and all lights shall be lit. The switch shall then be turned in the counter-clockwise direction and all panel lights shall dim and go out when the switch is turned all the way down. Any sample failing to conform to the applicable requirement shall be classed defective.

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4.4.5 Press-to-test Switch Lighting. - Twenty-eight, minus four, volts direct current (28.0 - 4.0 v.d.c.) shall be applied with the positive (+) polarity to the "+28 V Aircraft power Pin D in the rear panel connector PIRP of dwg. 9287814. Each switch having a press-to-light test capability shall be pressed in and the pilot light shall light. Any switch failing to light when pressed shall be classed defective.

4.4.6 Firing Test, Mines-Pairs. - After the control panel has completed and successfully passed the vibration tests of 4.4.1, power shall be applied to the rear panel connector PIRP. Twenty-eight minus four (28.0 -4.0) v.d.c. shall be applied with the proper polarity between the "+28 V Aircraft Power", Pin-D and "Aircraft Ground," pin - M of dwg. 9287814. Electronic counters and storage capability oscilloscopes shall be used to measure the required output requirements at the specified output pins when the control panel is tested as follows:

Test No. 1:

a) Set switch S2FP, "Quantity" in position 'A".
b) Set switch S6FP, "Interval" in position "1".
c) Set switch S3FP in "MINES-PAIRS" position.
d) Set switch SIFP in "SAFE" position.
e) Press switch S4FP, "FIRE" button in.

No output shall be recorded at any of the specified connector pins in the rear panel connector PIRP of dwg. 9287814.

Test No. 2:

a) Set switch SIFP In the "STANDBY" (READY) position and leave all other switches *in* the same position as stated in Test No. 1.
b) Press "FIRE" button, switch S4FP.

No output shall be recorded at any of the specified connector pins in the rear panel connector, PIRP of dwg. 9287814.

Test No. 3:

a) Leave switches S2FP, S3FP and S6FP in the same position as stated in Test No. 1.
b) Bring switch SIFP back into the "SAFE" position, then back thru "STANDBY" (READY) and into "ARM".
c) Press "FIRE" button, switch S4FP.

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The output pulses, as required by Curve A and Tables A, B, and D of dwg. 9288645 shall be measured for this position. These output pulses shall be recorded at connector Pin - E (left solenoid step), Pin - L (right solenoid (step), Pin - P (left canister fire) and Pin - R (right canister fire) in the rear panel connector, PIRP of dwg. 9287814. Solenoid outputs shall be measured across the circuit of dwg. 9278652. Canisters output shall be measured across a 7-ohm (7.0) resistive load.

Test No. 4:

a) Leave all switches except SIFP and S6FP in the same position as stated in Test No. 1.
b) Move switch SIFP back into the “SAFE” position, then back thru "STANDBY" (READY) Into “ARM”.
c) Move the “INTERVAL” switch S6FP into the next position.
d) Press "FIRE" button, switch S4FP.

The output shall be measured and comply with the requirements as specified in Test No. 3.

Test No. 5:

a) Repeat steps a and b of Test No. 4 and move the “INTERVAL” switch S6FP Into each sequential position and test that position by pressing the “FIRE” button, switch S4FP.
b) Measure the output pulses required by Curve A and Tables A, B and D of dwg. 9288645 for each of the “INTERVAL” positions. These output pulses shall be recorded at the connector pins as specified in Test No. 3.

Test No. 6:

a) Leave switch S3FP In “MINES-PAIRS” position.
b) Set switch S6FP, “INTERVAL” in position “8”.
c) Set “QUANTITY” switch, S2FP in position “B”.
d) Move switch SIFP back into the “SAFE” position, then back thru "STANDBY" (READY) into “ARM”.
e) Press "FIRE" button, switch S4FP.

Measure the output pulses required by Curve A and Tables A, B and D of dwg. 9288645 for this position. These output pulses shall be recorded at the connector pins as specified in Test No. 3.

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Test No. 7:

a) Repeat procedures a, b and d of Test No. 6.
b) Move "QUANTITY" switch, S2FP into each sequential position in the same manner and test that position by pressing the "FIRE" button, switch S4FP.
c) Measure the output pulses required by Curve A and Table A, B, and D of dwg. 9288645 for each of the “QUANTITY” positions. These output pulses shall be recorded at the connector pins as specified in Test No. 3.

4.4.7 Firing Test, Mines - Singles. - After the control panel has compled and successfully passed the firing test of 4.4.6, the d.c. power shall remain on the counters and storage oscilloscopes shall be used to measure the required outputs shown on Curve B, at the specified connector pins when the control panel Is tested as follows:

Test No. 1:

a) Set switch S2FP, "QUANTITY" in position “A”.
b) Set switch S6FP, “INTERVAL” in position “1”.
c) Set switch S3FP in “MINES-SINGLES’’- position.
d) Set switch SIFP in ‘SAFE” position.
e) Press switch S4FP, "FIRE" button In.

No output shall be recorded at any of the specified connector pins, in the rear panel connector, PIRP of dwg. 9287814.

Test No. 2:

a) Set switch SIFP in the “STANDBY” (READY) position and leave all other switches in the same position as stated in Test No. 1.
b) Press “FIRE” button, switch S4FP.

No output shall be recorded at any of the specified connector pins in the rear panel connector, PIRP of dwg. 9287814.

Test No. 3:

a) Leave switches, S2FP, S3FP and S6FP in the same position as stated In Test No. 1.
b) Bring switch SIFP back into the “SAFE” position, then back thru ”STANDBY” (READY) and into “ARM”.
c) press "FIRE" button, switch S4FP.

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The output pulses as required by Curve B and Tables A, B and D of dwg. 9288645 shall be measured for this position. These output pulses shall be recorded at the connector pins as specified in "Item 4.4.6, Test No. 3".

Test No. 4

a) Leave switch S3FP in "MINES-SINGLES" position.
b) Set switch S6FP, "INTERVAL" in position "8",
c) Move the "QUANTITY" switch, S2FP into the next position.
d) Move switch SIFP back into the "SAFE" position, then back thru "STANDBY" (READY) into "ARM".
e) Press "FIRE" button, switch S4FP.

The output pulses shall be measured and comply with the requirements of Curve B, and Tables A, B and D of dwg. 9288645. These output pulses shall be recorded at the connector pins as specified in "Item 4.4.6, Test No. 3".

Test No. 5:

a) Repeat steps a, b and d of Test No. 4 and move the "QUANTITY" switch, S6FP into each sequential position and test that position by pressing the "FIRE" button, switch S4F.
b) Measure the output pulses required by Curve B and Tables A, B and D of dwg. 9288645 for each of the "QUANTITY" positions. These output pulses shall be recorded at the connector pins as specified in "Item 4.4.6, Test No. 3".

4,4.8 Firing Test, Markers - Pairs. - Install Jumper JP3 on the printed wiring board "C" assembly per 9287765, Note -10 for this test. After the control panel has completed and successfully passed the firing test of 4.4.7, the d.c. power shall remain on the counters and storage oscilloscopes shall be used to measure the required outputs shown on Curve C at the specified connector pins when the control panel is tested as follows:

Test No. 1:

a) Set switch S2FP, "QUANTITY" in position "A".
b) Set switch S6FP, "INTERVAL" in position "1".
c) Set switch S3FP in "MARKERS-PAIRS" position.
d) Set switch SIFP in "SAFE" position.
e) Press switch S4FP, "FIRE" button in.

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No output shall be recorded at any of the specified output connector pins in the rear panel connector, PIRP of dwg. 9287814.

Test No. 2:

a) Set switch SIFP In the “STANDBY” (READY) position and leave all other switches in the same position as stated in Test No. 1.
b) Press “FIRE” button, switch S4FP.

No output shall be recorded at any of the specified connector pins in the rear panel connector, PIRP of dwg. 9287814.

Test No. 3:

a) Leave switches S2FP, S3FP and S6FP in the same position as stated in Test No. 1.
b) Bring switch SIFP back into the "SAFE" position, then back thru “STANDBY” (READY) and into "ARM".
c) Press “FIRE” button, switch S4FP.

The output pulses as required by Curve C and Tables A, B and D of dwg. 9288645 shall be measured for this position. These output pulses shall be recorded at the connector pins as specified in “Item 4.4.6, Test NO. 3“.

Test No. 4:

a) Leave switch S3FP in "MARKERS-PAIRS" position.
b) Set switch S6FP, “INTERVAL” in position “8”.
c) Move the "QUANTITY" switch, S2FP into the next position.
d) Move switch SIFP back into the ‘SAFE” position, then back thru “STANDBY” (READY) into “ARM”.
e) Press “FIRE” button, switch S4FP.

The output pulses shall be measured and comply with the requirements of Curve C, and Tables A, B and D of dwg. 9288645. These output pulses shall be recorded at the connector pins as specified in “Item 4.4.6, Test No. 3".

Test No. 5:

a) Repeat steps a, b and d of Test No. 4 and move the “QUANTITY” switch, S6FP into each sequential position and test that position by pressing the “FIRE” button, switch S4FP.

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b) Measure the output pulses required by Curve C and Tables A, B and D of dwg. 9288645 for each of the "QUANTITY" position. These output pulses shall be recorded at the connector pins as specified in "Item 4.4.6, Test No. 3".

4.4.9 Firing Test, Markers - Singles. - Install Jumper JP3 on the printed wiring board "C" assembly per 9287765, Note - 10 for this test. After the control panel has completed and successfully passed the firing test of 4.4.8, the d.c. power shall remain on the counters and storage oscilloscopes shall be used to measure the required outputs shown on Curve No. D at the specified connector pins when the control panel is tested as follows:

Test No. 1:

a) Set switch S2FP, "QUANTITY" In position "A".
b) Set switch S6FP, "INTERVAL" in position "l".
c) Set switch S3FP in "MARKERS-SINGLES" position.
d) Set switch SIFP in "SAFE" position.
e) Press switch S4FP, "FIRE" button in.

No output shall be recorded at any of the specified output connector pins in the rear panel connector, P1RP of dwg. 9287814.

Test No. 2:

a) Set switch SIFP in the "STANDBY" (READY) position and leave all other switches In the same position as stated in Test NO. 1.
b) Press "FIRE" button, switch S4FP.

No output shall be recorded at any of the specified connector pins in the rear panel connector, PIRP of dwg. 9287814.

Test No. 3:

a) Leave switches S2FP, S3FP and S6FP In the same position as stated in Test No. 1.
b) Bring switch SIFP back into the "SAFE" position, then back thru "STANDBY" (READY) and into "ARM".
c) Press "FIRE" button, switch S4FP.

The output pulses as required by Curve D and Tables A, B and D of dwg. 9288645 shall be measured for this position. These output pulses shall be recorded at the connector pins as specified in "Item 4.4.6, Test No. 3".

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Test No. 4:

a) Leave switch S3FP in “MARKERS-SINGLES” position.
b) Set switch S6FP, “INTERVAL” in position “8”.
c) Move the “QUANTITY” switch, S2FP into the next position.
d) Move switch SIFP back into the “SAFE” position, then back thru "STANDBY" (READY) Into “ARM”.
e. Press "FIRE" buttons switch S4FP.

The output pulses shall be measured and comply with the requirements of Curve D and Tables A, B and D of dwg. 9288645. These output pulses shall be recorded at the connector pins as specified In “Item 4.4.6, Test No. 3”.

Test No. 5:

a) Repeat steps a, b and d of Test No. 4 and move the “QUANTITY” switch, S6FP Into each sequential position and test that position by pressing the “FIRE” button switch S4Fp9
b) Measure the output pulses required by Curve D and Tables A, B and D of dug. 9288645 for each of the “QUANTITY” positions. These output pulses shall be recorded at the connector pins as specified in “Item 4.4.6, Test No. 3“.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level A. - The control panel assembly shall be preserved and packaged in accordance with dwg. 9287747.

5.2 Packing

5.2.1 Level A. - The control panel shall be packed in accordance with dwg.s 9287747 and CFR 49 parts 0-190.

5.3 Marking. - Marking of the carton shall comply with level A, dwg. 9287747.

5.4 Shipping. - When components of more than one lot. are shipped, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not speci-

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fied and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which Is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-10000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 In accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of results of contractor conducted examinations and tests (see 3.11). - The contractor shall send his results to:

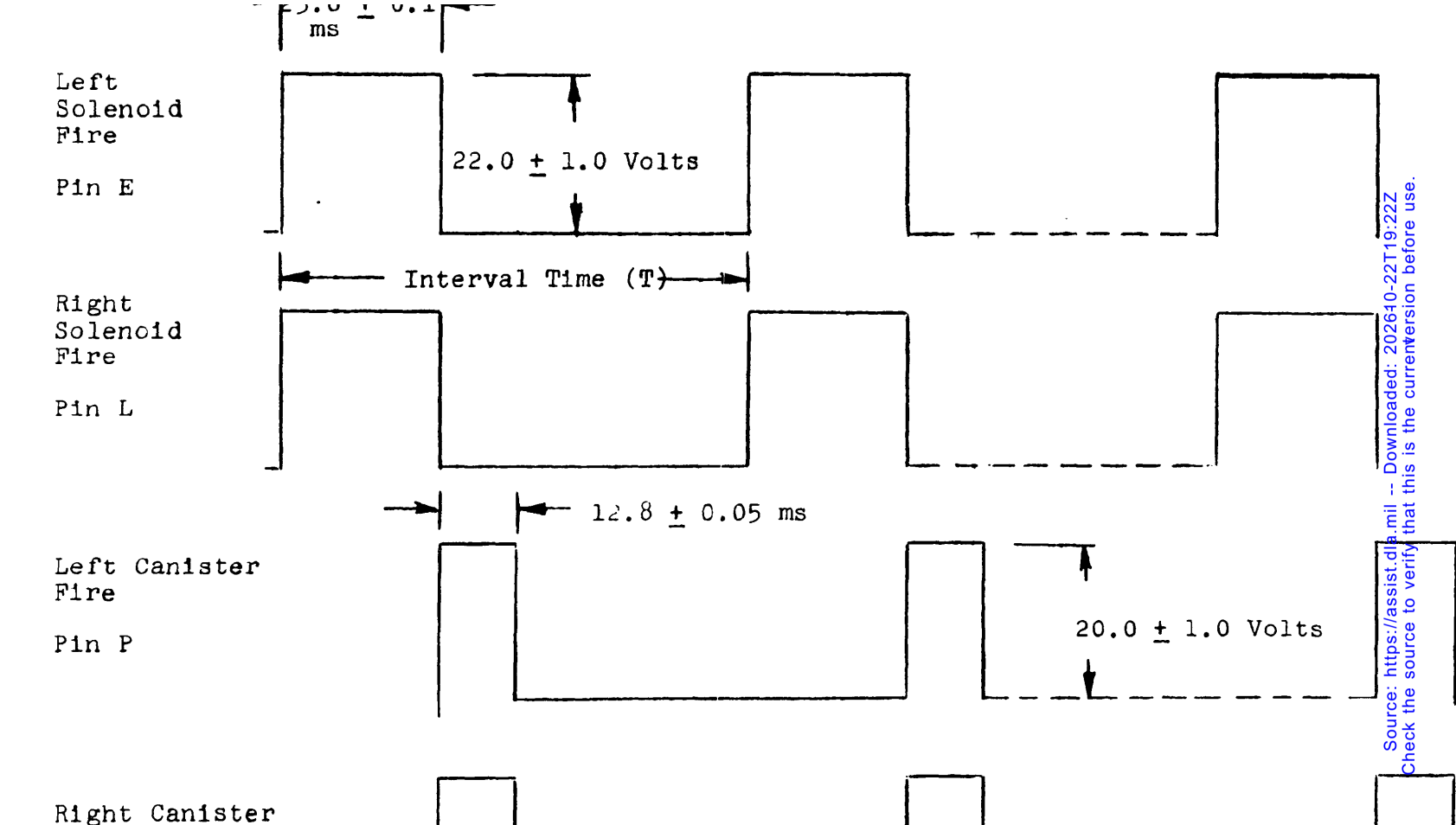
Commander:
Picatinny Arsenal
ATTN : SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-054

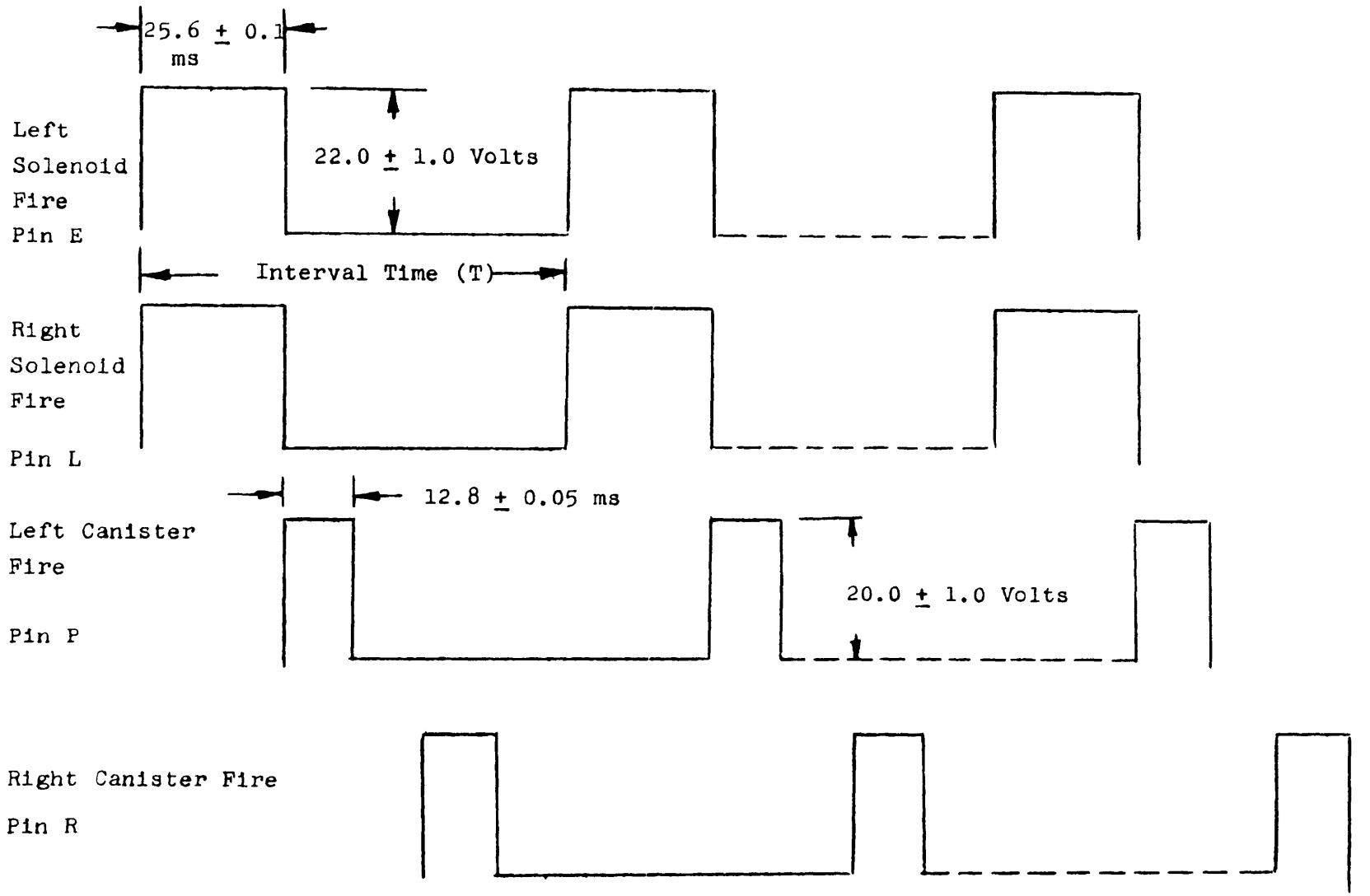
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CURVE A

MINES - PAIRS

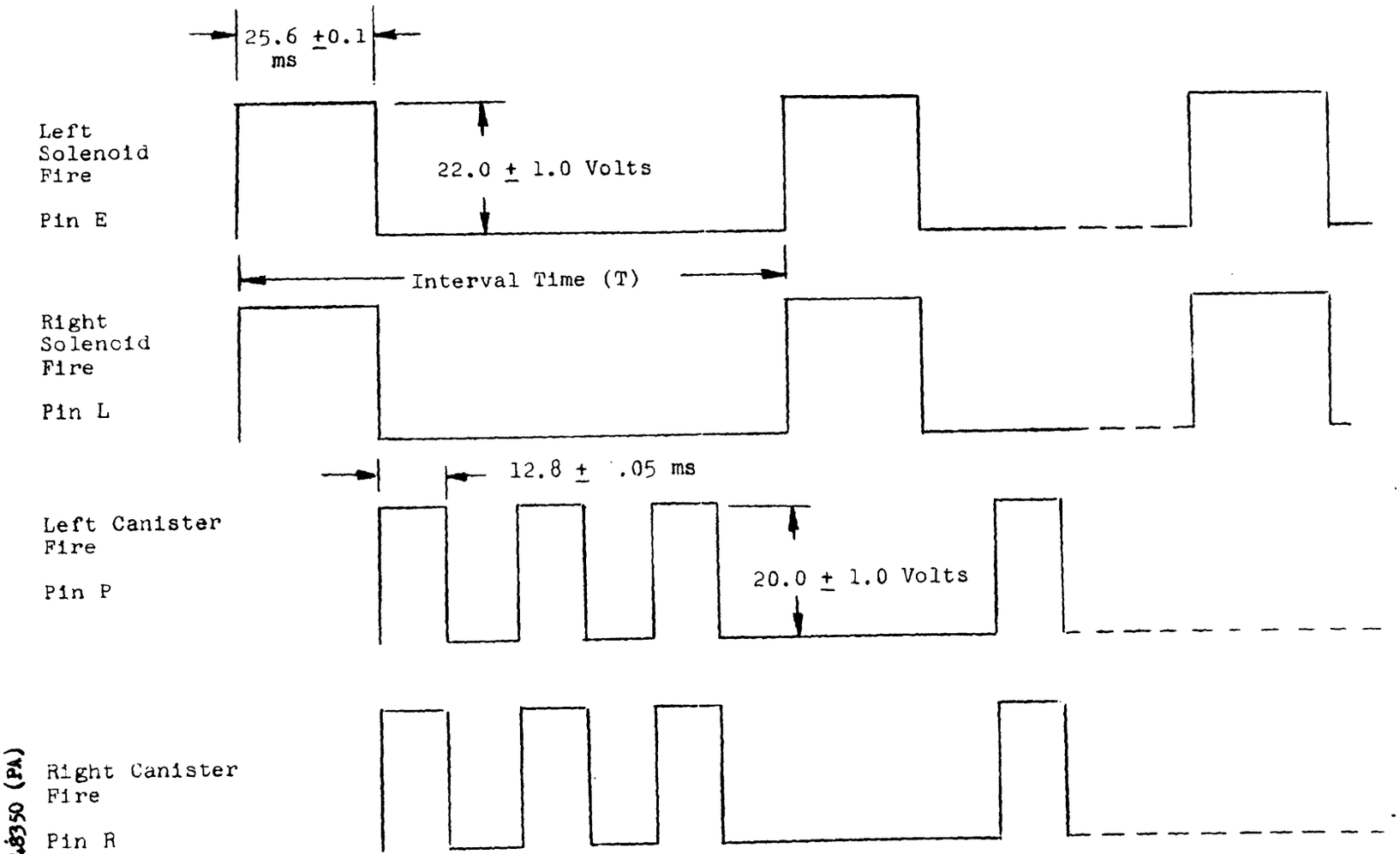
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CURVE B

MINES - SINGLES

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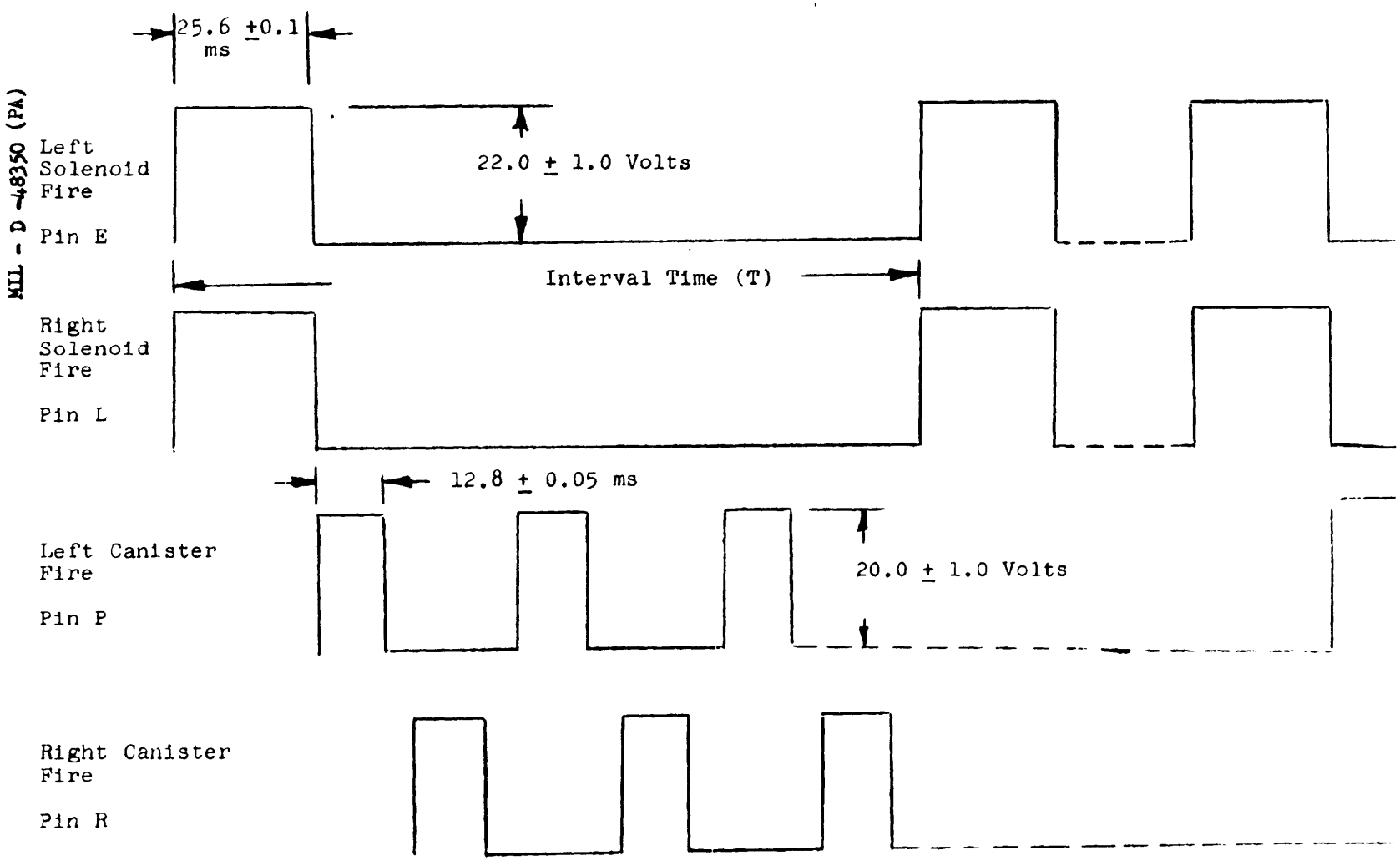


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CURVE C

MARKERS - PAIRS

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CURVE D

MARKERS - SINGLES

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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DEPARTMENT OF THE ARMY
DOD – 314



COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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